Project Report: HR Employee Attrition Prediction System

1. Data Understanding Report

1.1 Problem Statement

The objective of this project is to build a machine learning model that predicts whether an employee is likely to leave (attrite) the organisation.

Business Context

Employee attrition is a costly challenge for organisations. High turnover leads to:

- Increased recruitment and onboarding costs
- Loss of institutional knowledge and productivity
- Reduced team morale and continuity

A predictive system can help HR teams:

- Identify at-risk employees early
- Understand the key drivers of attrition
- Take targeted retention actions before employees resign

1.2 Objective

- Predict employee attrition (Yes / No)
- Identify key factors contributing to attrition
- Provide actionable recommendations to the HR department

1.3 Target Variable

- Attrition
  - 1 → Attrition (Yes – employee left)
  - 0 → No Attrition (No – employee stayed)

1.4 Input Variables

The dataset contains 35 columns. After removing constant/ID columns, 31 features are used. Key input variables include:

Numerical Features:

- Age
- DailyRate
- DistanceFromHome
- HourlyRate

- MonthlyIncome
- MonthlyRate
- NumCompaniesWorked
- PercentSalaryHike
- TotalWorkingYears
- TrainingTimesLastYear
- YearsAtCompany
- YearsInCurrentRole
- YearsSinceLastPromotion
- YearsWithCurrManager

Categorical Features:

- BusinessTravel
- Department
- EducationField
- Gender
- JobRole
- MaritalStatus
- OverTime

Ordinal / Satisfaction Ratings (scale 1–4):

- Education
- EnvironmentSatisfaction
- JobInvolvement
- JobLevel
- JobSatisfaction
- PerformanceRating
- RelationshipSatisfaction
- StockOptionLevel
- WorkLifeBalance

1.5 Initial Data Checks

The following checks were performed on load:

- df.shape → Dataset dimensions: 1,470 rows × 35 columns
- df.info() → Data types and null value summary
- df.describe() → Statistical summary (mean, std, min, max, quartiles)
- df.head() / df.tail() → Sample row inspection

Observations

- No null / missing values detected in the original dataset
- Mix of numerical, ordinal, and categorical (object) variables
- Target variable is imbalanced: 84% No Attrition · 16% Yes Attrition

- Four columns contained constant values and were removed immediately

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1.6 Data Cleaning Summary

Constant / Useless Columns Removed

Column	Reason for Removal
EmployeeNumber	Unique identifier – no predictive value
EmployeeCount	Constant value (always 1)
Over18	Constant value (always 'Y')
StandardHours	Constant value (always 80)

Null Value Handling

Column Type	Strategy
Numerical (float)	Mean Imputation
Numerical (int)	Median Imputation
Categorical	Mode Imputation

Note: No null values were present in this dataset; imputation logic was implemented as a defensive best-practice pattern.

Duplicate Handling

- Checked using df.duplicated().sum()
- Removed using df.drop_duplicates()
- Result: 0 duplicate rows found

Error Value Handling

- Scanned all categorical columns for values like "?", "@", "#", "NA", "N/A", "none"
- Replaced any found error values with NaN, then re-applied imputation
- Result: No error values detected in this dataset

Column Segregation

- Categorical columns identified via: df.select_dtypes(include='object').columns
- Numerical columns identified via: df.select_dtypes(exclude='object').columns

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1.7 Outlier Detection & Treatment

Technique Used: IQR Method

Applied to all numerical columns:

- Q1 = df.quantile(0.25) | Q3 = df.quantile(0.75)
- IQR = Q3 − Q1

- Lower Bound (LB) = Q1 − 1.5 × IQR
- Upper Bound (UB) = Q3 + 1.5 × IQR

Columns with Outliers Detected

Column	Action
MonthlyIncome	Outliers removed beyond IQR bounds
DailyRate	Outliers removed beyond IQR bounds
TotalWorkingYears	Outliers removed beyond IQR bounds
YearsAtCompany	Outliers removed beyond IQR bounds
YearsSinceLastPromotion	Outliers removed beyond IQR bounds
NumCompaniesWorked	Outliers removed beyond IQR bounds
DistanceFromHome	Outliers removed beyond IQR bounds
PercentSalaryHike	Outliers removed beyond IQR bounds
YearsWithCurrManager	Outliers removed beyond IQR bounds
YearsInCurrentRole	Outliers removed beyond IQR bounds

Action

- Outlier rows removed from dataset (IQR hard cut)
- Boxplots saved to output_plots/01_outlier_boxplots.png for visual verification

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1.8 Exploratory Analysis Summary

Target Distribution

- 84% of employees: No Attrition | 16% of employees: Yes Attrition
- Clear class imbalance – handled via class_weight='balanced' in all models
- Plot saved: output_plots/02_target_distribution.png

Univariate Analysis

Performed on all major numerical and categorical columns:

- Histograms plotted for: Age, MonthlyIncome, TotalWorkingYears, YearsAtCompany, DistanceFromHome, YearsSinceLastPromotion
- Bar charts plotted for: Department, JobRole, MaritalStatus, BusinessTravel, Gender, OverTime
- Plots saved: output_plots/03_univariate_numerical.png | output_plots/04_univariate_categorical.png

Bivariate Analysis

Analysed how each feature relates to the Attrition target variable:

- Attrition rate by Department, MaritalStatus, BusinessTravel, Gender, OverTime, JobRole
- Numerical feature distributions split by Attrition: Yes vs No (overlaid histograms)
- Plots saved: output_plots/05_bivariate_categorical_vs_attrition.png | output_plots/06_bivariate_numerical_vs_attrition.png

Correlation Heatmap

- Heatmap of all numerical feature correlations produced
- Plot saved: output_plots/07_correlation_heatmap.png

Key Observations

- OverTime has a strong association with attrition – employees working overtime leave at higher rates
- Single / unmarried employees show significantly higher attrition rates
- Frequent business travellers have elevated attrition compared to non-travellers
- Lower MonthlyIncome correlates with higher attrition probability
- Younger employees (< 35 years) and those with fewer TotalWorkingYears leave more frequently
- Employees in Sales Representative and Laboratory Technician roles show highest attrition rates

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2. Feature Engineering Documentation

2.1 Target Encoding

Column	Original Values	Encoded Values	Method
Attrition	Yes / No	1 / 0	Manual map()

2.2 Categorical Encoding

Different encoding strategies were applied based on the nature of each categorical column:

Techniques Used

Type	Columns	Method
Binary categorical	Gender, OverTime	Manual map() (Male=1/Female=0, Yes=1/No=0)
Ordinal	BusinessTravel	Ordinal Encoding (Non-Travel=0, Travel_Rarely=1, Travel_Frequently=2)
Nominal	Department, EducationField, JobRole, MaritalStatus	One-Hot Encoding (pd.get_dummies, drop_first=True)

Note: drop_first=True in One-Hot Encoding eliminates the dummy variable trap and reduces multicollinearity.

2.3 Feature Transformation

- StandardScaler applied to all input features after train-test split
- Scaler fitted only on training data (X_train) to prevent data leakage into the test set

- Both X_train_scaled and X_test_scaled produced for model input

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2.4 Feature Selection

Removed

Column	Reason
EmployeeNumber	Unique ID – no predictive signal
EmployeeCount	Constant (all rows = 1)
Over18	Constant (all rows = 'Y')
StandardHours	Constant (all rows = 80)

No additional feature selection (e.g., VIF, RFE) was performed beyond removing constants, as Random Forest feature importance was used post-training to rank remaining features.

2.5 Final Feature Set

After all preprocessing steps:

- Input matrix X: shape depends on dataset after outlier removal, approximately 1,200+ rows × 31+ columns (after one-hot expansion)
- Target vector y: Attrition (0 = No, 1 = Yes)
- Train set: 80% of data | Test set: 20% of data
- Stratify=y ensures both splits maintain the same 84:16 class ratio

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3. Model Development and Evaluation Report

3.1 Dataset Splitting

Split	Size	Variables
Training set	80%	X_train, y_train
Test set	20%	X_test, y_test

- train_test_split used with random_state=42 for reproducibility
- stratify=y applied to preserve class distribution in both splits

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3.2 Models Implemented

1. Logistic Regression

- Baseline binary classification model
- Interpretable coefficients – shows direction and magnitude of feature influence
- class_weight='balanced' to handle 84:16 class imbalance
- max_iter=1000 to ensure full convergence

2. Decision Tree

- Handles non-linear relationships with simple if/else rules
- Highly explainable – decision paths can be shown to non-technical stakeholders
- max_depth=20, class_weight='balanced'

3. Random Forest

- Ensemble of 100 decision trees (n_estimators=100)
- Reduces overfitting through bagging and feature randomisation
- max_depth=10, class_weight='balanced'
- Primary source of feature importance scores

4. K-Nearest Neighbours (KNN)

- Non-parametric model – classifies based on similarity to nearest training examples
- n_neighbors=7 (tuned via GridSearchCV)
- No explicit class_weight parameter – handles imbalance indirectly via scaling

5. Naive Bayes

- Fast probabilistic classifier based on Bayes' theorem
- Assumes feature independence (Gaussian distribution over numerics)
- Useful as a fast baseline comparison model

3.3 Model Training

- All models trained using: model.fit(X_train_scaled, y_train)
- StandardScaler applied before training to normalise all features to the same scale
- Prevents distance-based models (KNN) from being biased by feature magnitude differences

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3.4 Cross Validation

- 5-Fold Stratified Cross-Validation applied to all models on the training set
- StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
- Metrics evaluated per fold: Accuracy, Precision, Recall, F1-Score
- Mean scores reported across all 5 folds for reliable performance estimation

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3.5 Hyperparameter Tuning

GridSearchCV was applied to find optimal hyperparameters for key models:

Model	Parameter Tuned	Values Tried
Logistic Regression	C (regularisation)	0.01, 0.1, 1, 10, 100
Decision Tree	max_depth	5, 10, 15, 20
Random Forest	n_estimators	50, 100, 200
KNN	n_neighbors	3, 5, 7, 9, 11

3.6 Model Evaluation Metrics

Metrics Used (Classification)

Metric	Description
Accuracy	Overall percentage of correct predictions
Precision	Of all predicted attritions, how many were actual?
Recall	Of all actual attritions, how many did the model catch?
F1-Score	Harmonic mean of Precision and Recall (primary metric)
Confusion Matrix	TP / FP / TN / FN breakdown
Misclassification Rate	Percentage of incorrect predictions (1 − Accuracy)

F1-Score is the primary selection metric because the dataset is imbalanced and both false positives (unnecessary interventions) and false negatives (missed attrition) carry business cost.

3.7 Results Summary

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	76%	37%	68%	48%
Decision Tree	76%	34%	53%	42%
Random Forest	83%	47%	46%	47%
KNN	81%	31%	10%	15%
Naive Bayes	62%	25%	70%	37%

3.8 Model Selection

Selected Model:

- Best model is selected automatically by the script based on highest F1-Score on the test set.
- In representative runs, Logistic Regression and Random Forest produce comparable F1 scores, with Logistic Regression frequently selected due to consistent balanced performance.

Reason for Selection Criteria

- Highest F1-Score on the held-out test set

- Better balance of Precision and Recall for the imbalanced class
- Good generalisation – low gap between CV score and test score
- Interpretable enough to support HR business recommendations

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4. Feature Importance Analysis

4.1 Method

Feature importances extracted from the trained Random Forest model using rf.feature_importances_ (mean decrease in impurity across all trees).

Top 15 features visualised and saved to output_plots/08_feature_importance.png.

4.2 Top Features Driving Attrition

Rank	Feature	Importance Score	Interpretation
1	TotalWorkingYears	0.0877	Less experienced employees leave more
2	OverTime	0.0844	Overtime is a major attrition trigger
3	Age	0.0758	Younger employees have higher attrition
4	MonthlyIncome	0.0706	Lower salary correlates with leaving
5	YearsWithCurrManager	0.0650	Short tenure with manager signals risk
6	YearsAtCompany	0.0583	Newer hires are more likely to leave
7	StockOptionLevel	0.0417	No stock options reduces loyalty
8	YearsInCurrentRole	0.0407	Stagnant roles drive attrition
9	DistanceFromHome	0.0375	Long commutes increase attrition risk
10	JobSatisfaction	0.0346	Low satisfaction predicts departure

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5. Insights and Recommendations Report

5.1 Key Insights

- OverTime is the second most important predictor – employees required to work overtime are at significantly higher risk of leaving the organisation
- Lower MonthlyIncome and no salary hike increases attrition probability substantially

- Single/unmarried employees leave at a higher rate – likely due to greater mobility and fewer financial obligations
- Employees who travel frequently for business show higher attrition than non-travellers
- Short tenure with current manager (YearsWithCurrManager) is an early warning signal
- Sales Representatives and Laboratory Technicians are the roles with highest attrition rates
- Low JobSatisfaction and low JobInvolvement scores are leading indicators of future attrition
- Employees with no StockOptionLevel have significantly higher departure rates

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5.2 Business Recommendations

1. Reduce Overtime

- OverTime is the second most important attrition driver

2. Improve Salary and Compensation

- MonthlyIncome is the 4th most important feature

3. Manager Quality and Engagement

- Short YearsWithCurrManager is a key risk indicator

4. Job Enrichment for At-Risk Roles

- Sales Representatives and Laboratory Technicians show the highest attrition

5. Address Work-Life Balance for Travellers

- Frequent travellers show elevated attrition – cap non-essential travel

6. Focus on Early Tenure Employees

- Employees in their first 1–3 years at the company are most at risk

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5.3 Model Usage

Step	Description
Input	Applicant / employee feature data (all 31 input columns after preprocessing)
Process	Run through trained best model (selected by F1-Score at runtime)
Output	Attrition prediction: 0 = No Attrition | 1 = Attrition (with probability %)
Use Case	Monthly HR risk scoring, exit interview targeting, retention programme prioritisation

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6. Conclusion

This project developed an end-to-end HR Employee Attrition Prediction system using IBM HR Analytics data (1,470 employees, 35 features). The workflow included data preparation, analysis, feature engineering, model training, hyperparameter tuning, and evaluation. Five classification models were compared using F1-Score to address class imbalance, with the best model automatically selected. Key insights identified factors such as Overtime, Salary, Manager Tenure, and Job Role as major drivers of attrition, helping HR teams improve retention and reduce turnover costs.